

Soham Deshpande

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Objective

Motivated AI & ML postgraduate with a solid foundation in Python, SQL, and full-stack development (Django, MySQL, REST APIs). Hands-on experience building AI-powered applications, backend systems, and data pipelines, with strong analytical and communication skills. Interested in software development, data analysis, and research roles.

Projects

1. InfraNova AI — Thermal Infrared to RGB Satellite Image Translation

- Developed a Pix2Pix conditional GAN (PyTorch) to translate Landsat 9 Thermal Infrared (TIR) imagery into RGB imagery, enabling improved visual interpretation of thermal satellite data.
- Integrated YOLOv8 with the generated RGB imagery to perform automated road-network extraction and evaluate downstream object-detection performance on translated images.
- Built an end-to-end training and inference pipeline featuring Automatic Mixed Precision (AMP), custom loss functions, deterministic checkpointing, and dataset validation.
- Designed a U-Net generator with PatchGAN discriminator and combined loss (adversarial, L1, perceptual, SSIM, chroma) for realism and color fidelity.
- Built a FastAPI + React web app for interactive inference and trained the model for 226 epochs on dual Tesla T4 GPUs (Kaggle) using DataParallel and AMP.

2. Translytic — AI-Based Real-Time Video Captioning & Translation System

- Developed a real-time video captioning and translation system using Python, OpenAI Whisper, Gemini API, Google Translate API, edge-tts, OpenCV, and Tkinter.
- Built an end-to-end AI pipeline for speech-to-text, translation, subtitle generation, and multilingual dubbing with timestamp synchronization.
- Integrated REST APIs and JSON-based data processing with automatic fallback mechanisms for transcription and translation.
- Implemented multithreaded video playback, frame buffering, and audio-video synchronization for smooth real-time performance.

3. NoteFlow — Full Stack Notes Management Platform

- Developed a full-stack notes management application using Django, MySQL, HTML, CSS, and JavaScript.
- Implemented user authentication, CRUD operations, category-based filtering, and role-based access control.
- Designed relational database schemas using Django ORM and MySQL with support for per-user notes and dashboard analytics.
- Built AJAX-based pin functionality and real-time search capabilities without full page refreshes.

Education

Master of Computer Applications (AI & ML)
Ramdeobaba University, Nagpur | 2025 – Present

Bachelor of Computer Applications
G.H. Rasoni College of Engineering and Management, Nagpur | 2022 – 2025

12th Grade (HSC)
Pratibha Junior College (Maharashtra State Board) | 2022

10th Grade (SSC)
S.N.B.P. International School (CBSE) | 2020

Skills

Technical Skills

- Programming: Python, Java, SQL
- AI/ML & Data: NumPy, Pandas, OpenCV, Matplotlib, PyTorch, Scikit-learn
- AI Assistants: ChatGPT, Claude, Gemini
- Backend & APIs: Django, FastAPI, REST APIs, HTML, CSS
- Data & Databases: MySQL, Oracle Apex, Data Cleaning, Data Validation
- Cloud & Dev Tools: Git, GitHub, VS Code, Google Colab, Kaggle
- Data Visualization: Power BI, Excel

Soft Skills:

- Analytical Thinking | Research & Documentation | Problem Solving | Communication | Team Collaboration

Certifications

- Google Cloud Career Launchpad – Computing Foundations Track, Google Cloud (Apr 2026)
- Assets, Threats, and Vulnerabilities – Google Cybersecurity Certificate, Coursera (Apr 2026)
- Connect and Protect: Networks and Network Security – Google Cybersecurity Certificate, Coursera (Mar 2026)
- Tools of the Trade: Linux and SQL – Google Cybersecurity Certificate, Coursera (Apr 2026)